

Combinational

classmate

Date _____

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Assignment 3 .

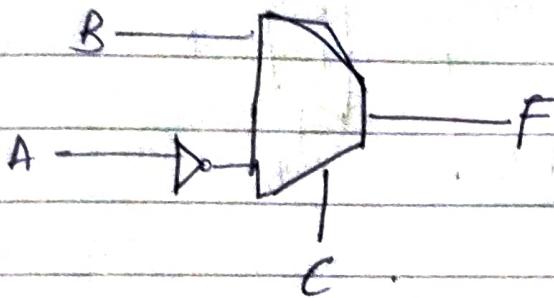
1. Implement the following functions using
(i) 2:1 MUX & Not gates
(ii) 4:1 MUX

(a) $F(A, B, C) = \bar{A}C + B\bar{C}$

i) 2:1 MUX & NOT

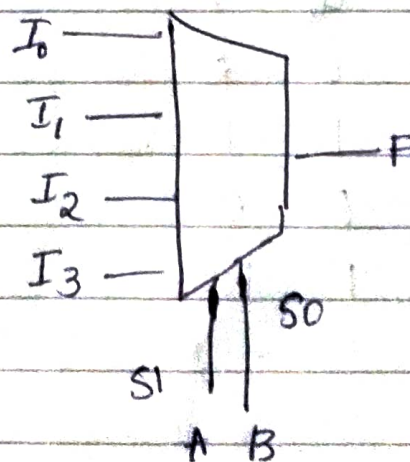
When $C = 0$, $F = B$

$C = 1$ $F = \bar{A}$



(i') 4:1 MUX .

A	B	F
0	0	C
0	1	1
1	0	0
1	1	$\bar{C}$



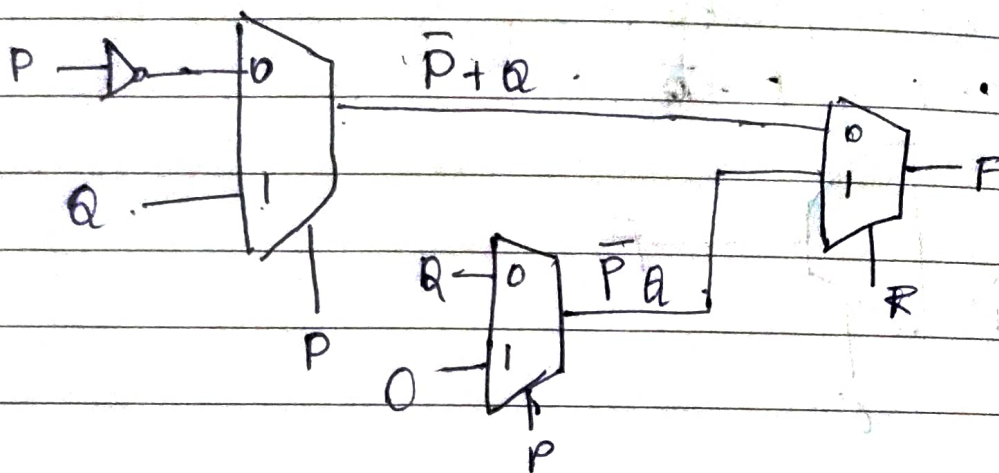
b) $F(P, Q, R) = \overline{P}\overline{R} + QR + \overline{P}Q$

(i) Using 2:1 Mux

When $R=0$ $F = \overline{P} + Q + \overline{P}Q = \overline{P} + Q$

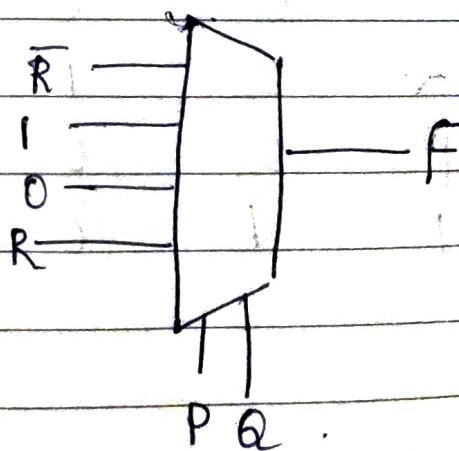
$R=1$ $F = \overline{P}Q$

$I_0 = \overline{P}Q$ $I_1 = \overline{P}Q$



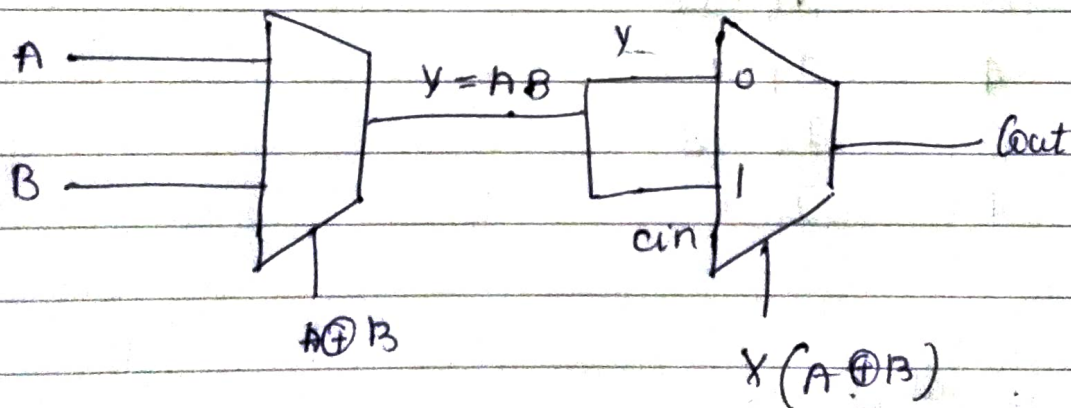
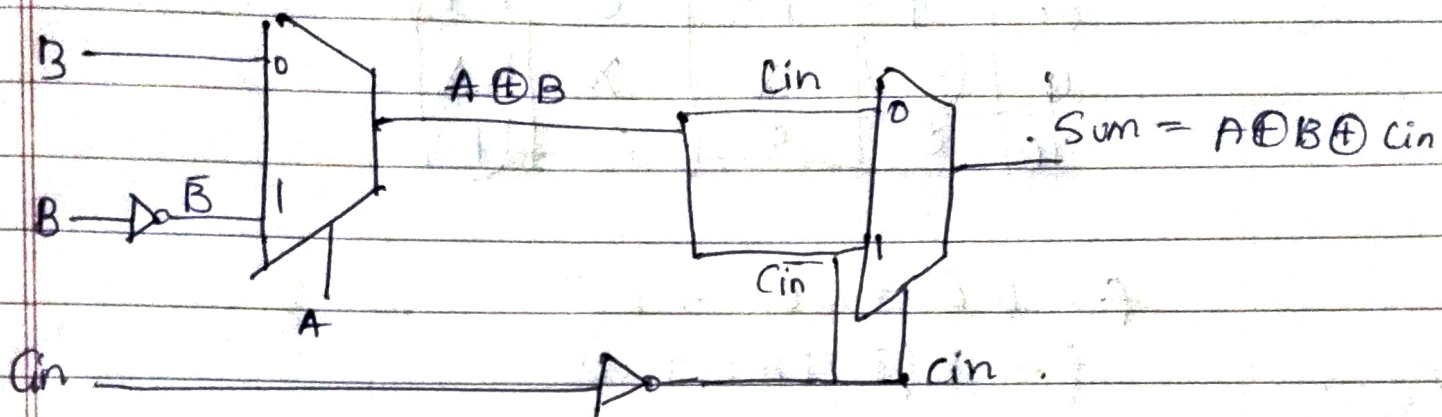
(ii) Using 4:1 Mux

P	Q	F
0	0	$\overline{R}$
0	1	1
1	0	0
1	1	$\overline{R}$



2) Implement the functionality of a full adder using 2:1 Muxes only

Full adder: $Sum = A \oplus B \oplus C$
 $Carry = A \cdot B + C_{in}(A \oplus B)$



$$A \oplus B = \bar{A}B + A\bar{B} \quad \text{When } A=1 \quad o/p = \bar{B}$$

$$A=0 \quad o/p = B$$

Q3. Simplify $f(a, b, c, d) = \sum m(2, 4, 5, 6, 10) + \sum d(12, 13, 14, 15)$ & implement the SOP expression using 4:1 Mux.

		cd			
		00	01	11	10
ab	00	0	0	0	1
	01	1	1	0	1
	11	X	X	X	X
	10	0	0	0	1

$$f = b\bar{c} + c\bar{d} + b\bar{d}$$

$$f = b(\bar{c} + \bar{d}) + c\bar{d}$$

a	b	Mux i/p
0	0	$I_0 = \bar{c}\bar{d}$
0	1	$I_1 = 1$
1	0	$I_2 = c\bar{d}$
1	1	$I_3 = 1$

